MGS 657

Homework 4

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Creating and Running ETL Pipelines

Executive summary:

During this assignment I learned a lot about oracles autonomous AI databases and also ran into some errors. Before starting this assignment, I did have some previous experience with databases, Oracle databases, SQL, creating a virtual machine on Oracle for big data, and querying throughout other courses at The University at Buffalo. I have had no experience with ETL pipelines and Apache Hop though. This assignment helped me learn many new things about the ETL process and insert/updating records into databases.

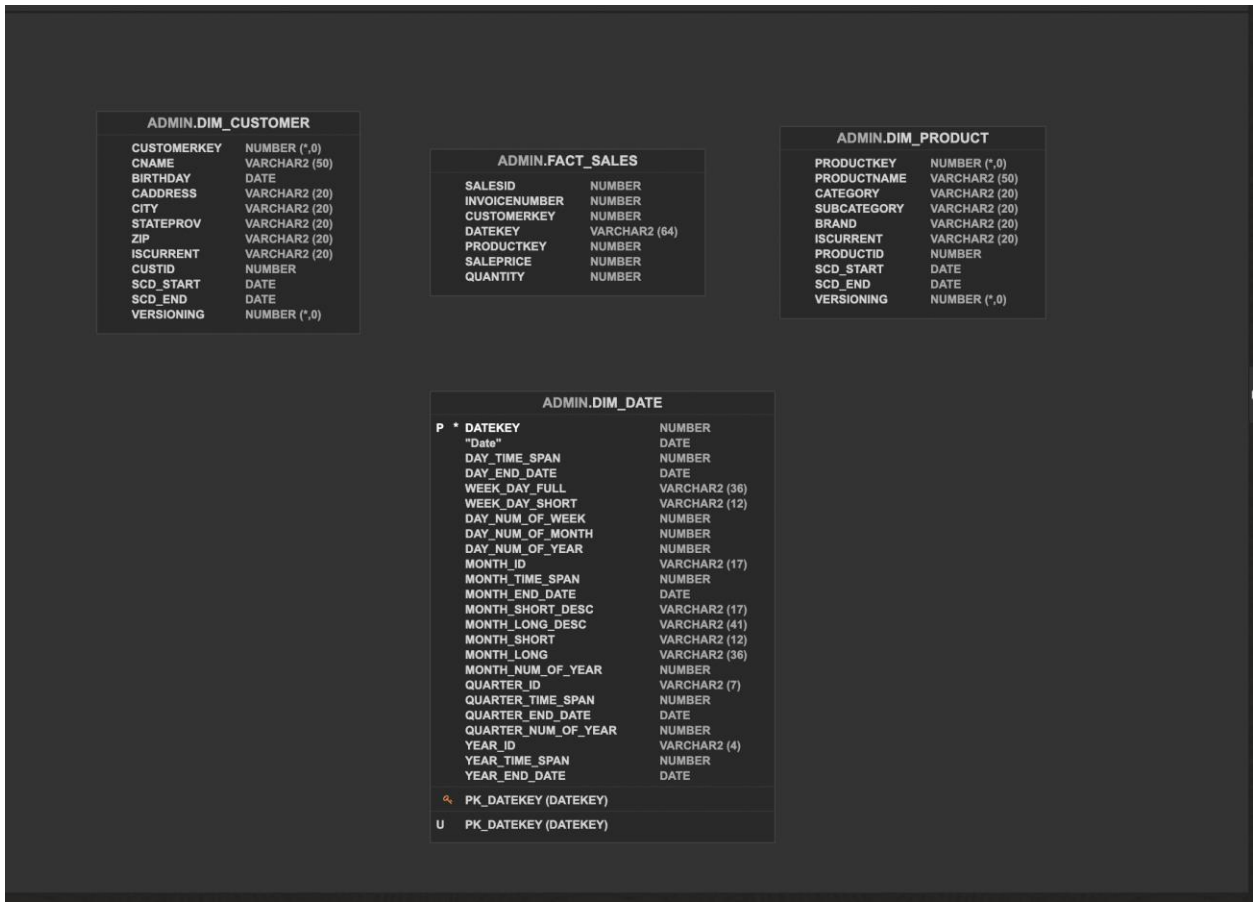
One issue I ran into was when inserting new rows into the tables for the IsCurrent is that it would use a 0 and 1, instead of Y and N. When creating the product and customer inputs for the fact table, I had to use the SQL of "SELECT * FROM ADMIN.DIM_CUSTOMER WHERE **TO_CHAR(ISCURRENT) IN ('Y', '1')**" and "SELECT * FROM ADMIN.DIM_PRODUCT WHERE **TO_CHAR(ISCURRENT) IN ('Y', '1')**", instead of SELECT * FROM ADMIN.DIM_PRODUCT **WHERE ISCURRENT='Y'**".

Another issue I ran into was when entering the manual connection URL. I was using ChatGPT to help me with the URL by removing the retry count and delay and adding the service name to my URL. When using the URL ChatGPT provided, I got an error message

and after inspecting I realized that ChatGPT misspelled the service name by switching an a and an e. This made me realize how important it is to double check the work of AI.

In summary, this assignment helped me learn a lot about ETL pipelines and how the knowledge of configuring them beforehand is important to the overall process.

Diagram of the database:

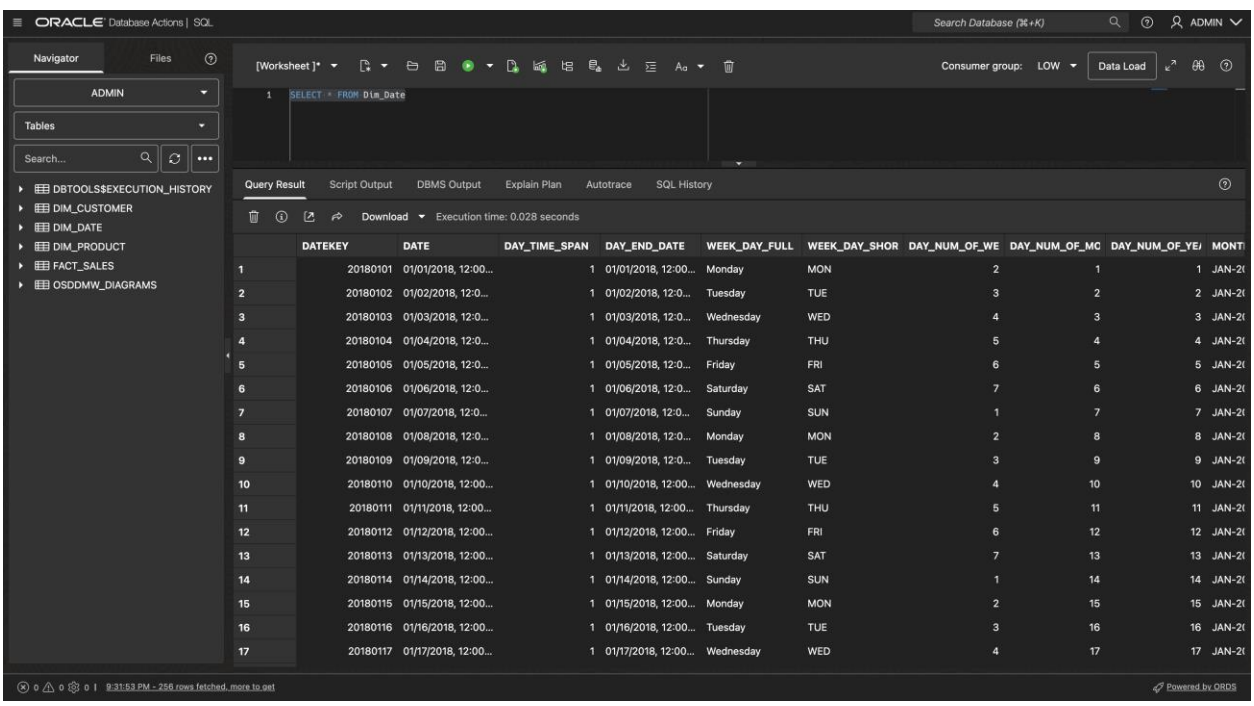


Answer the following questions in your report: Based on the diagram generated, what is this database missing that you'd expect to see? Why might it be missing this component? Use materials we've discussed in class and research (citing sources) to write no more than 2-4 sentences in response:

There is no foreign key relationships defined in the Oracle database. From the class lecture notes, the Kimball method harps on the importance of star schemas relying on FK relationships to enforce dimensional structure. According to a stack overflow comment, “Reasons not to use Foreign Keys: you are making the DB work extra on every CRUD operation because it has to check FK consistency. This can be a big cost if you have a lot of churn”. Not having set FK relationships make it easier for the pipelines to run.

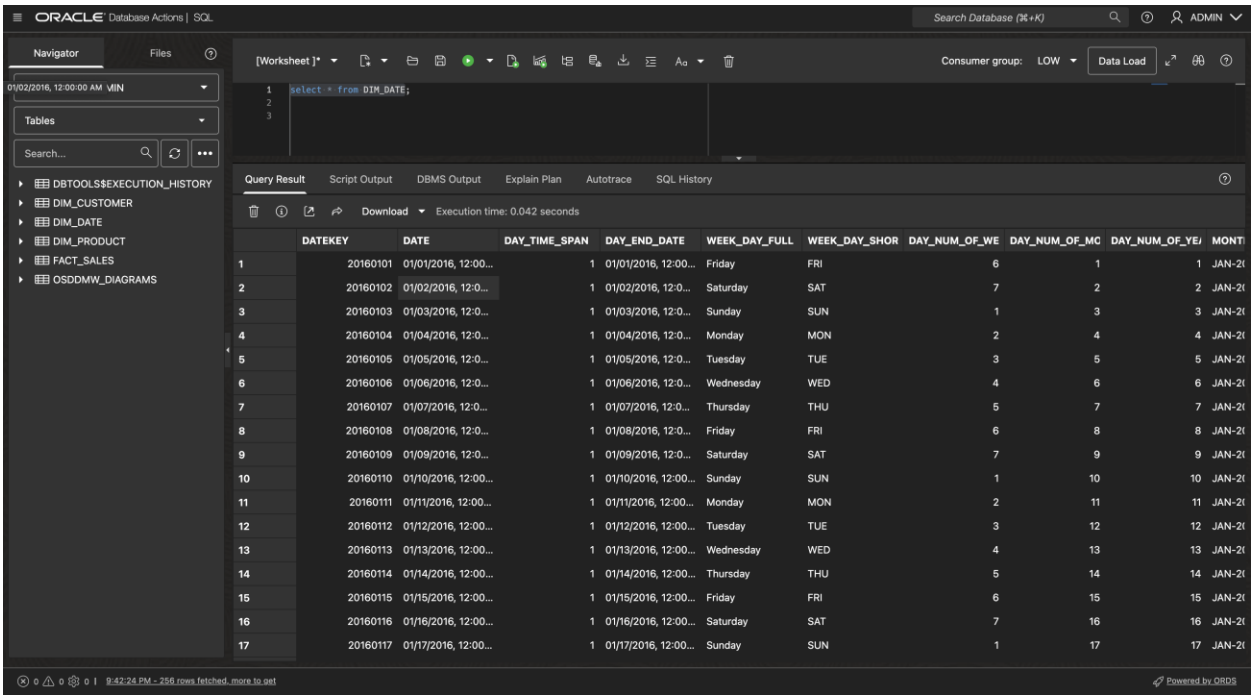
Ed Lucas. “What’s wrong with foreign keys?” *Stack Overflow*, 17 Sept. 2008, https://stackoverflow.com/questions/83147/whats-wrong-with-foreign-keys.

SELECT * FROM Dim_Date:



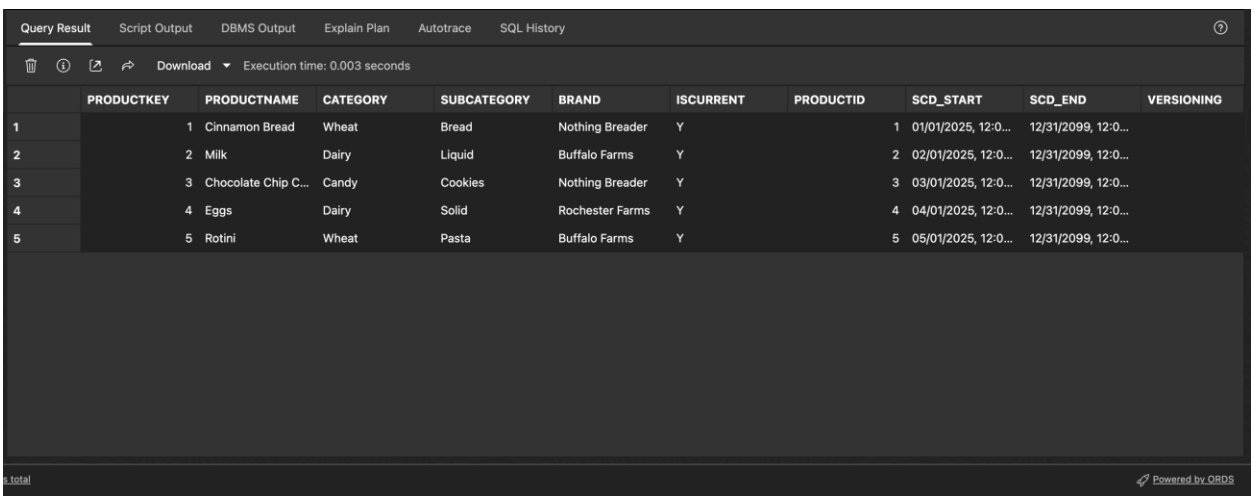
	DATEKEY	DATE	DAY_TIME_SPAN	DAY_END_DATE	WEEK_DAY_FULL	WEEK_DAY_SHOR	DAY_NUM_OF_WE	DAY_NUM_OF_MC	DAY_NUM_OF_YE/	MONT
1	20180101	01/01/2018, 12:00...	1	01/01/2018, 12:0...	Monday	MON	2	1	1	JAN-2(
2	20180102	01/02/2018, 12:0...	1	01/02/2018, 12:0...	Tuesday	TUE	3	2	2	JAN-2(
3	20180103	01/03/2018, 12:0...	1	01/03/2018, 12:0...	Wednesday	WED	4	3	3	JAN-2(
4	20180104	01/04/2018, 12:0...	1	01/04/2018, 12:0...	Thursday	THU	5	4	4	JAN-2(
5	20180105	01/05/2018, 12:0...	1	01/05/2018, 12:0...	Friday	FRI	6	5	5	JAN-2(
6	20180106	01/06/2018, 12:0...	1	01/06/2018, 12:0...	Saturday	SAT	7	6	6	JAN-2(
7	20180107	01/07/2018, 12:0...	1	01/07/2018, 12:0...	Sunday	SUN	1	7	7	JAN-2(
8	20180108	01/08/2018, 12:0...	1	01/08/2018, 12:0...	Monday	MON	2	8	8	JAN-2(
9	20180109	01/09/2018, 12:0...	1	01/09/2018, 12:0...	Tuesday	TUE	3	9	9	JAN-2(
10	20180110	01/10/2018, 12:00...	1	01/10/2018, 12:00...	Wednesday	WED	4	10	10	JAN-2(
11	20180111	01/11/2018, 12:00...	1	01/11/2018, 12:00...	Thursday	THU	5	11	11	JAN-2(
12	20180112	01/12/2018, 12:00...	1	01/12/2018, 12:00...	Friday	FRI	6	12	12	JAN-2(
13	20180113	01/13/2018, 12:00...	1	01/13/2018, 12:00...	Saturday	SAT	7	13	13	JAN-2(
14	20180114	01/14/2018, 12:00...	1	01/14/2018, 12:00...	Sunday	SUN	1	14	14	JAN-2(
15	20180115	01/15/2018, 12:00...	1	01/15/2018, 12:00...	Monday	MON	2	15	15	JAN-2(
16	20180116	01/16/2018, 12:00...	1	01/16/2018, 12:00...	Tuesday	TUE	3	16	16	JAN-2(
17	20180117	01/17/2018, 12:00...	1	01/17/2018, 12:00...	Wednesday	WED	4	17	17	JAN-2(

SELECT * FROM Dim_Date updated:



	DATEKEY	DATE	DAY_TIME_SPAN	DAY_END_DATE	WEEK_DAY_FULL	WEEK_DAY_SHOR	DAY_NUM_OF_WE	DAY_NUM_OF_MC	DAY_NUM_OF_YE/	MONT
1	20160101	01/01/2016, 12:00...	1	01/01/2016, 12:00...	Friday	FRI	6	1	1	JAN-2(
2	20160102	01/02/2016, 12:0...	1	01/02/2016, 12:0...	Saturday	SAT	7	2	2	JAN-2(
3	20160103	01/03/2016, 12:0...	1	01/03/2016, 12:0...	Sunday	SUN	1	3	3	JAN-2(
4	20160104	01/04/2016, 12:0...	1	01/04/2016, 12:0...	Monday	MON	2	4	4	JAN-2(
5	20160105	01/05/2016, 12:0...	1	01/05/2016, 12:0...	Tuesday	TUE	3	5	5	JAN-2(
6	20160106	01/06/2016, 12:0...	1	01/06/2016, 12:0...	Wednesday	WED	4	6	6	JAN-2(
7	20160107	01/07/2016, 12:0...	1	01/07/2016, 12:0...	Thursday	THU	5	7	7	JAN-2(
8	20160108	01/08/2016, 12:0...	1	01/08/2016, 12:0...	Friday	FRI	6	8	8	JAN-2(
9	20160109	01/09/2016, 12:0...	1	01/09/2016, 12:0...	Saturday	SAT	7	9	9	JAN-2(
10	20160110	01/10/2016, 12:00...	1	01/10/2016, 12:00...	Sunday	SUN	1	10	10	JAN-2(
11	20160111	01/11/2016, 12:00...	1	01/11/2016, 12:00...	Monday	MON	2	11	11	JAN-2(
12	20160112	01/12/2016, 12:00...	1	01/12/2016, 12:00...	Tuesday	TUE	3	12	12	JAN-2(
13	20160113	01/13/2016, 12:00...	1	01/13/2016, 12:00...	Wednesday	WED	4	13	13	JAN-2(
14	20160114	01/14/2016, 12:00...	1	01/14/2016, 12:00...	Thursday	THU	5	14	14	JAN-2(
15	20160115	01/15/2016, 12:00...	1	01/15/2016, 12:00...	Friday	FRI	6	15	15	JAN-2(
16	20160116	01/16/2016, 12:00...	1	01/16/2016, 12:00...	Saturday	SAT	7	16	16	JAN-2(
17	20160117	01/17/2016, 12:00...	1	01/17/2016, 12:00...	Sunday	SUN	1	17	17	JAN-2(

Select * from Dim_Product:



	PRODUCTKEY	PRODUCTNAME	CATEGORY	SUBCATEGORY	BRAND	ISCURRENT	PRODUCTID	SCD_START	SCD_END	VERSIONING
1	1	Cinnamon Bread	Wheat	Bread	Nothing Breader	Y	1	01/01/2025, 12:0...	12/31/2099, 12:0...	
2	2	Milk	Dairy	Liquid	Buffalo Farms	Y	2	02/01/2025, 12:0...	12/31/2099, 12:0...	
3	3	Chocolate Chip C...	Candy	Cookies	Nothing Breader	Y	3	03/01/2025, 12:0...	12/31/2099, 12:0...	
4	4	Eggs	Dairy	Solid	Rochester Farms	Y	4	04/01/2025, 12:0...	12/31/2099, 12:0...	
5	5	Rotini	Wheat	Pasta	Buffalo Farms	Y	5	05/01/2025, 12:0...	12/31/2099, 12:0...	

SELECT * FROM Dim_Product 2:

Query Result | Script Output | DBMS Output | Explain Plan | Autotrace | SQL History

Download Execution time: 0.003 seconds

	PRODUCTKEY	PRODUCTNAME	CATEGORY	SUBCATEGORY	BRAND	ISCURRENT	PRODUCTID	SCD_START	SCD_END	VERSIONING
1	1	Cinnamon Bread ...	Wheat	Bread	Nothing Breader	Y	1	01/01/2025, 12:0...	12/31/2099, 12:0...	
2	2	Milk	Dairy	Liquid	Buffalo Farms	Y	2	02/01/2025, 12:0...	12/31/2099, 12:0...	
3	3	Chocolate Chip C...	Candy	Cookies	Nothing Breader	Y	3	03/01/2025, 12:0...	12/31/2099, 12:0...	
4	4	Eggs	Dairy	Solid	Rochester Farms	0	4	04/01/2025, 12:0...	11/07/2025, 11:59...	
5	5	Rotini	Wheat	Pasta	Buffalo Farms	Y	5	05/01/2025, 12:0...	12/31/2099, 12:0...	
6	102	Eggs	Poultry	Solid	Rochester Farms	1	4	11/07/2025, 11:59...	12/31/2199, 11:59...	
7	103	Sugary Cereal	Wheat	Cereal	Food For You	1	6	11/07/2025, 11:59...	12/31/2199, 11:59...	

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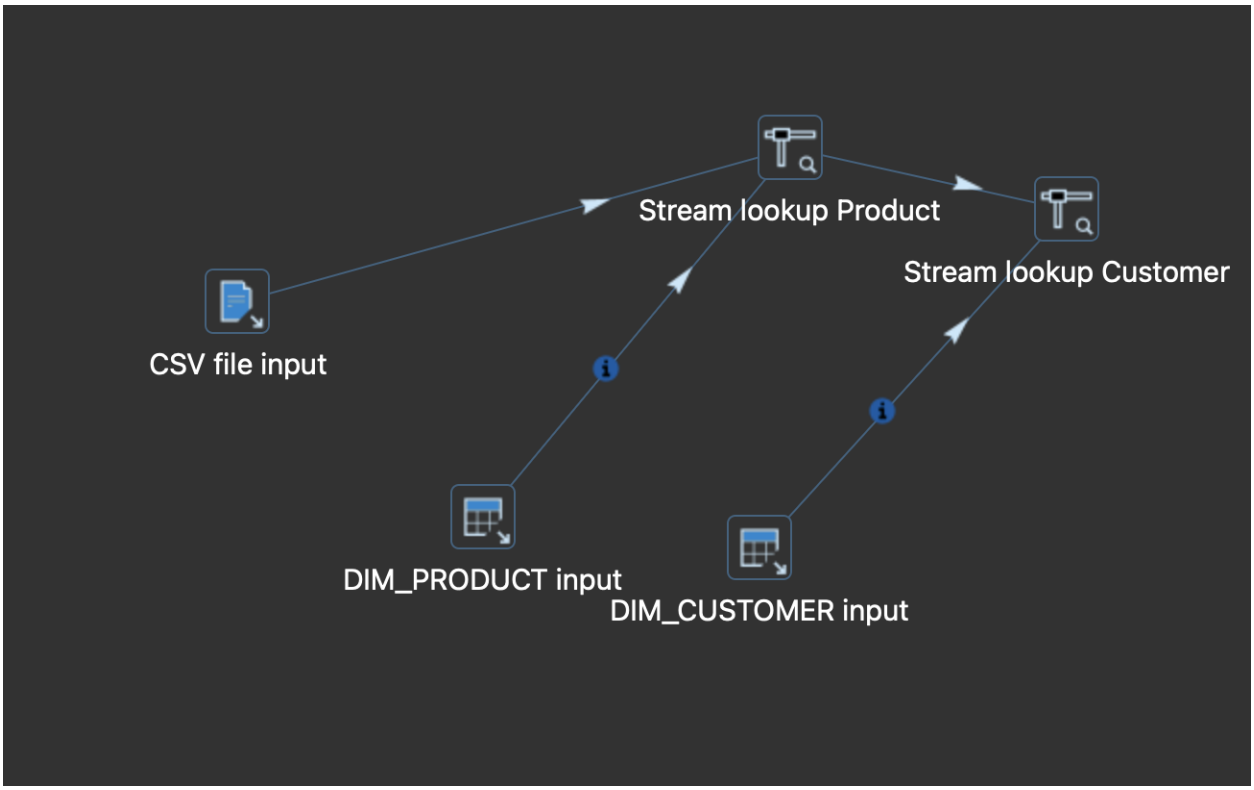
Apahe Hop Pipeline:



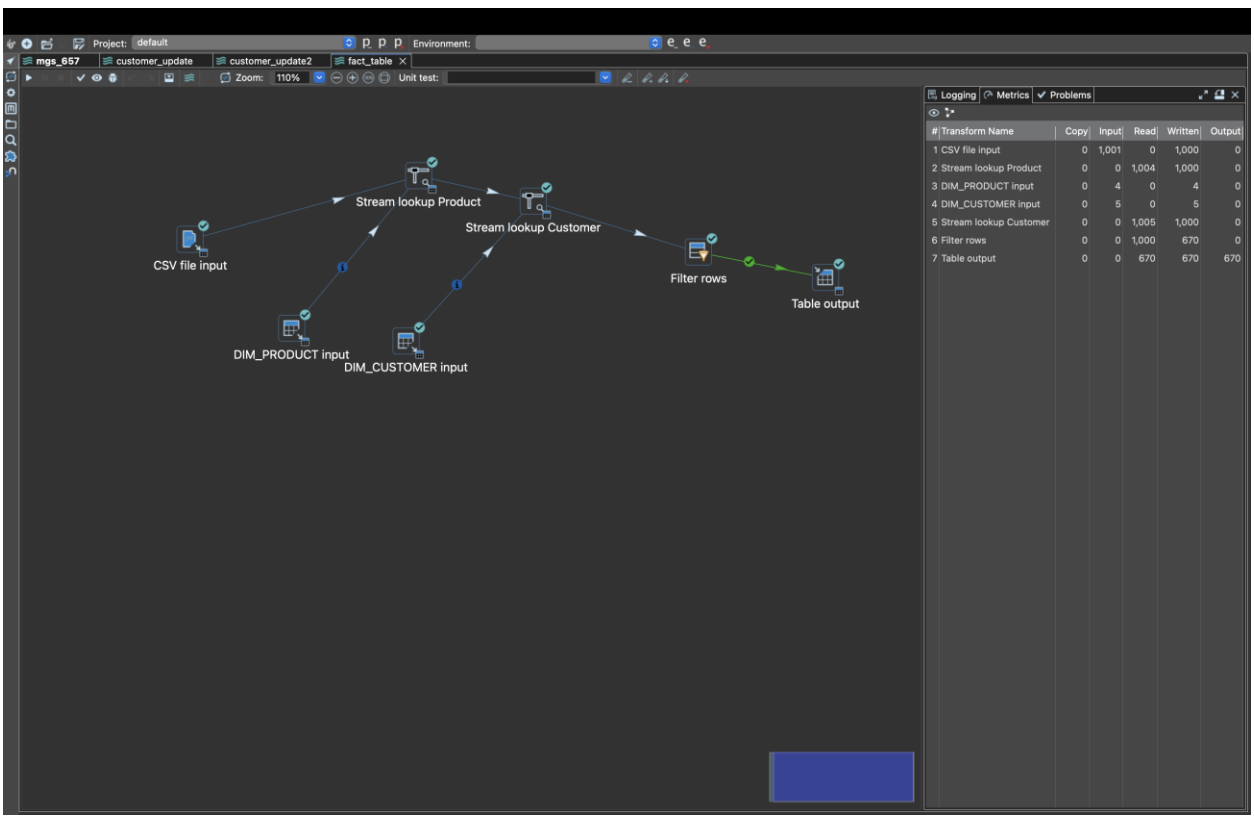
SELECT * FROM Dim_Customer:

Query Result | Script Output | DBMS Output | Explain Plan | Autotrace | SQL History

Download | Execution time: 0.001 seconds

	STOMERKEY	CNAME	BIRTHDAY	CADDRESS	CITY	STATEPROV	ZIP	ISCURRENT	CUSTID	SCD_START
1	1009	Jeep Jeeperson	(null)	123 Cool St.	Buffalo	NY	14043	1	2	11/07/2025, 11:45...
2	1	Dominic Sellitto	01/01/1956, 12:0...	123 ABC St.	Buffalo	NY	14222	0	1	12/31/2021, 12:00...
3	2	Jeep Jeeperson	02/02/1979, 12:0...	123 Cool St.	Buffalo	NY	14222	0	2	12/31/2021, 12:00...
4	3	Sally Sallerson	03/03/1989, 12:0...	415 Awesome Pl.	Rochester	NY	54321	Y	3	12/31/2021, 12:00...
5	1008	Dominic Sellitto	(null)	123 New St.	Rochester	NY	14321	1	1	11/07/2025, 11:45...
6	1010	James Bond	(null)	543 Bond Rd.	Buffalo	NY	14222	1	4	11/07/2025, 11:45...
7	1011	Jennifer Lopez	(null)	91 Perfect Ave.	Rochester	NY	14321	1	5	11/07/2025, 11:45...

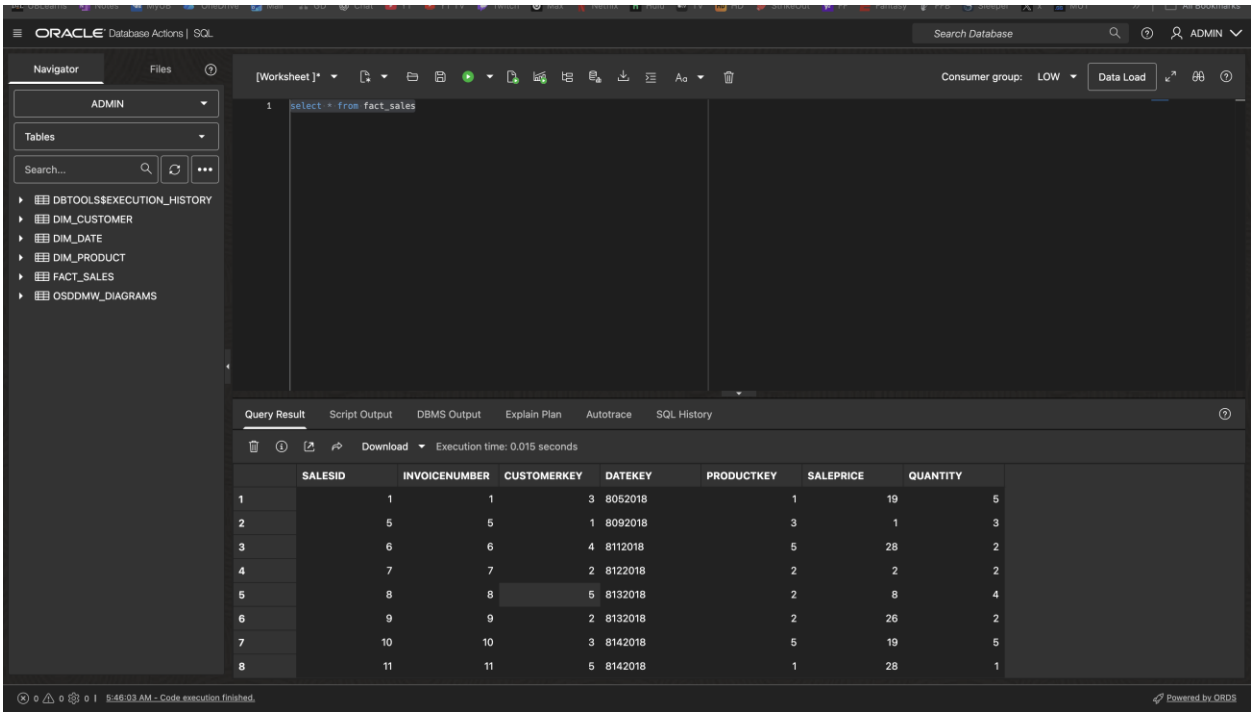
Entire Apahe Hop Pipeline:



You might have noticed we're not doing a lookup for the date dimension. Write 1-2 sentences detailing why we don't need to.

The fact table already contains the date key values that align with the primary keys from the date dimension we already created and inserted the data from the csv. The ETL can load the date key into the fact table without a separate lookup.

SELECT * FROM Fact_Sales:



	SALESID	INVOICENUMBER	CUSTOMERKEY	DATEKEY	PRODUCTKEY	SALEPRICE	QUANTITY
1	1	1	3	8052018	1	19	5
2	5	5	1	8092018	3	1	3
3	6	6	4	8112018	5	28	2
4	7	7	2	8122018	2	2	2
5	8	8	5	8132018	2	8	4
6	9	9	2	8132018	2	26	2
7	10	10	3	8142018	5	19	5
8	11	11	5	8142018	1	28	1